(Peter) Yi Wei

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Research Interests

I work on algebraic geometry and arithmetic geometry. More specifically, I'm interested in

- Geometry of syzygies of algebraic varieties
- Brill-Noether theory of curves
- Moduli theory of curves and K3 surfaces

Employment

University of Arkansas — Fayetteville, Arkansas, USA
Visiting Assistant Professor in Mathematics — August 2025 - now

Education

University of Wisconsin-Madison — Madison, Wisconsin, USA
Ph. D. in Mathematics — September 2019 - June 2025
Thesis: Brill-Noether theory of curves and syzygies of canonical curves
Thesis Advisor: Michael Kemeny

St Edmund Hall, University of Oxford — Oxford, UK
Visiting Student Program — October 2018- March 2019

University of Chinese Academy of Sciences (UCAS) — Beijing, CHINA
B. Sc. in Mathematics — September 2015 - July 2019
Dissertation: Euler Characteristic of Theta Divisors for Spectral Curves.
Advisor: Baohua Fu

Preprint & Publications

- Kemeny, Michael, and Wei, Peter Yi, *The Geometry Syzygy Conjecture in Positive Characteristics*, arXiv:2509.00844.
- Wei, Yi, *Generic Green's Conjecture and Generic Geometric Syzygy Conjecture in Positive Characteristic*, submitted to a journal. arXiv:2109.12187.
- with Li, A., Yin, X., Xu, B. et al, *Decoding topologically associating domains with ultra-low resolution Hi-C data by graph structural entropy* , Nature Communications **9**, 3265 (2018). https://doi.org/10.1038/s41467-018-05691-7

Awards & Grants

Research Assistantship — May - August 2024
UW-Madison

Research Assistantship — May - August 2023
UW-Madison

Exceptional Teaching Award — June 2022
For demonstrated excellence in teaching over four or more semesters — *UW-Madison*

Henry Schaerf Mathematics Graduate award — November, 2021
For demonstrated promise in the academic work — *UW-Madison*

Research Assistantship — July - August, 2021
UW–Madison, Wisconsin Alumni Research Foundation

Physical Sciences Award — 2019
UW-Madison, Graduate Studies

Selected Organization

Algebra Learning Seminar — Fall 2025
A weekly meeting series of algebra learning seminar — *University of Arkansas, Fayetteville*

Madison Math Circle — Fall 2024
A weekly series of mathematically based activities aimed at interested middle school and high school students. — *UW-Madison*

Directed Reading Program — Fall 2020 - Spring 2021
Pairing undergraduate students with graduate mentors for semester-long independent studies — *UW-Madison*

Invited Talks

Syzygies of canonical curves — September 2025
Algebra Seminar Fall 2025 — *University of Arkansas, Fayetteville*

Green's Conjecture on Syzygies of Canonical Curves — August 2023
Academy of Mathematics and Systems Science Chinese Academy of Sciences, China

Geometric Syzygy Conjecture in char p, with reveries from Ogus' result on a versal deformation of K3 surfaces — October 2021
Algebra and Algebraic Geometry Seminar Fall 2021 — *UW-Madison*

Expository Talks

Syzygies of adjoint linear series on algebraic curves — December 2023
Graduate Algebraic Geometry Seminar — *UW-Madison*

de Rham Cohomology in char p and Cartier Isomorphism — September 2023
Graduate Algebraic Geometry Seminar — *UW-Madison*

Brill-Noether Theory of Curves and K3 Surfaces — March 2023
Graduate Algebraic Geometry Seminar — *UW-Madison*

Pathologies in Algebraic Geometry — October 2021
Graduate Algebraic Geometry Seminar — *UW-Madison*

The S-Unit Equation: p-adic Approaches — September 2021
Graduate Student Number Theory/Representation Theory Seminar — *UW-Madison*

K3 Surfaces and Their Moduli — December 2020
Graduate Algebraic Geometry Seminar — *UW-Madison*

Moduli Problems for Elliptic Curves — November 2020
Seminar on Abelian Varieties — *UW-Madison*

Perfectoid Rings II — November 2020
Student Arithmetic Geometry Reading Seminar — *University of Minnesota*

Perfectoid Rings I — October 2020
Student Arithmetic Geometry Reading Seminar — *University of Minnesota*

Theorem of the Cube — October 2020
Seminar on Abelian Varieties — *UW-Madison*

Belyi's Theorem and Grothendieck's Dessins d'enfants — October 2020
Graduate Student Number Theory/Representation Theory Seminar — *UW-Madison*

An Introduction to ℓ-adic Galois Representations — April 2020
Graduate Student Number Theory/Representation Theory Seminar — *UW-Madison*

An Introduction to Grothendieck-Riemann-Roch — March 2020
Graduate Algebraic Geometry Seminar — *UW-Madison*

Teaching Experience

Survey of Calculus — Fall 2025
Instructor — *University of Arkansas, Fayetteville*

Math 221: Calculus and Analytic Geometry I — Spring & Fall 2024, Fall 2022
Teaching Assistant — *UW-Madison*

Math 222: Calculus and Analytic Geometry II — Fall 2023, Fall 2019
Teaching Assistant — *UW-Madison*

Math 320: Linear Algebra and Differential Equations Spring 2023
Teaching Assistant *UW-Madison*

Math 319: Techniques in Ordinary Differential Equations Summer 2022
Instructor *UW-Madison*

Math 319: Techniques in Ordinary Differential Equations Spring 2022
Teaching Assistant *UW-Madison*

Math 112: College Algebra Fall 2021
Lecturer *UW-Madison*

Math 234: Calculus - Functions of Several Variables Spring 2021
Teaching Assistant *UW-Madison*

Math 211: Calculus Fall 2020, Summer 2020
Teaching Assistant *UW-Madison*

Math 340: Elementary Matrix and Linear Algebra Spring 2020
Teaching Assistant *UW-Madison*

Conference Attended

Western Algebraic Geometry Symposium November 7 - 9, 2025
Poster Session *University of Oregon*

Workshop on Tevelev Degrees and Related Topics October 26 - 27, 2024
Participant *University of Illinois Urbana-Champaign, USA*

Syzygies and Hilbert Schemes September 24 - 28, 2024
Participant *Jagiellonian University, Krakow, Poland*

2nd Simons Math Summer Workshop: Moduli July 1 - 19, 2024
Participant *Simons Center for Geometry and Physics, Stony brook University, USA*

LMS Invited Lectures 2024 July 1 - 5, 2024
Participant *Imperial College London, UK*

Arizona Winter School 2024: Abelian Varieties March 1 - 6, 2024
Participant *University of Arizona in Tucson, USA*

Winter School on New Applications of Mixed Hodge Modules January 15 - 26, 2024
Participant *Simons Center for Geometry and Physics, Stony brook University, USA*

Western Algebraic Geometry Symposium (WAGS) November 4 - 5, 2023
Participant *Washington University in St. Louis, MO, USA*

AGNES @ UPenn 2023 October 13 - 15, 2023
Participant *University of Pennsylvania, Philadelphia, USA*

AGNES Summer School on Intersection Theory on Moduli Spaces July 11 - 14, 2023
Participant *Brown University, Providence, RI, USA*

Derived Categories, Moduli Spaces, and Counting Invariants July 3 - 7, 2023
Poster Session *Imperial College London, UK*

CMND 2023 Thematic Program on Rationality and Hyperbolicity June 19 - 30, 2023
Participant *University of Notre Dame, Indiana, USA*

Arithmetic, Birational Geometry, and Moduli Spaces June 12 - 16, 2023
Poster Session *Brown University, Providence, RI, USA*

Syzygies and Regularity April 14 - 16, 2023
Participant *University of Illinois at Chicago, USA*

Derived Categories, Moduli Spaces, and Hyperkähler Varieties August 1 - 5, 2022
Participant *University of Michigan at Ann Arbor, USA*

Geometric Structures (Re)United: Higgs Bundles, Geometric Structures, and Character Varieties June 13-17, 2022
Participant *University of Illinois at Chicago, USA*

Arizona Winter School 2022: Automorphic Forms Beyond GL_2 March 5 - 10, 2022
Participant *University of Arizona in Tucson, USA*

Arizona Winter School 2020: Nonabelian Chabauty March 7 - 11, 2020
Participant *University of Arizona in Tucson, USA*

Miscellaneous

Hobbies

Calligraphy 2003-present

- I learn how to write Chinese calligraphy since I was a child. I perform it excellently on the traditional context. I was president of *Calligraphy and Painting Club* in UCAS for academic year 2016-2017.

Tennis 2021 - present

- I am a passionate tennis player, aiming to play at a competitive level.

Piano 2021 - present

- I am a grown-up neophyte in music and in particular piano. I am passionate about classical music, especially in the Romantic Era.